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(54) **BACKLIGHT MODULE INCLUDING LIGHT GUIDE STRUCTURE**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,860,171 A * 8/1989 Kojima G02B 6/0041 362/616
10,935,715 B2 * 3/2021 Park G02B 6/005
(Continued)

FOREIGN PATENT DOCUMENTS

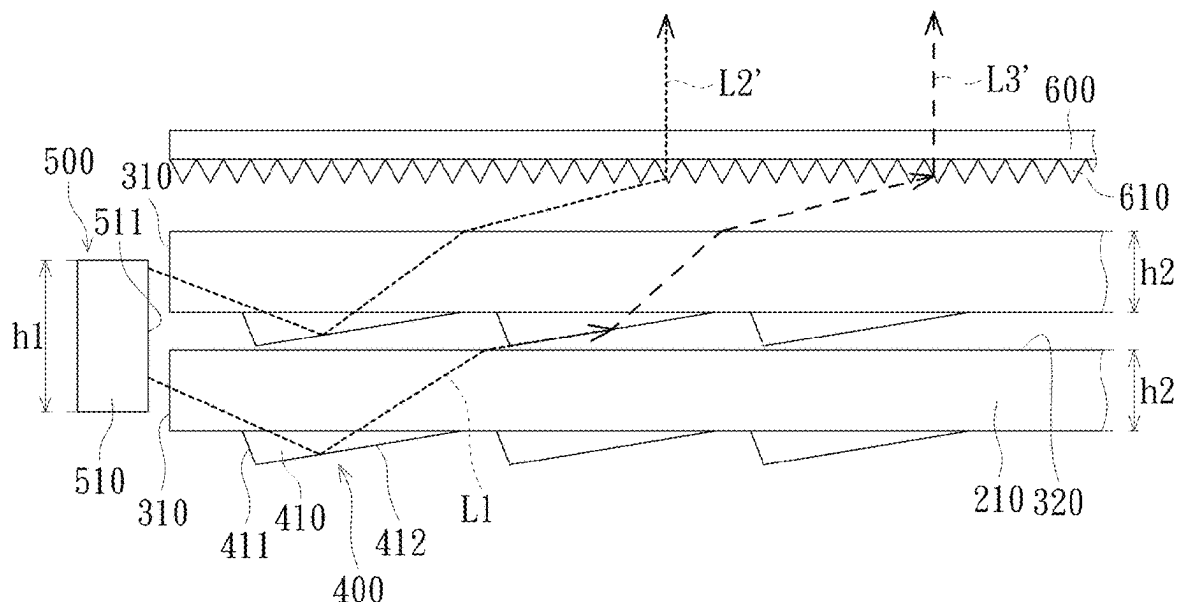
CN 113204070 A 8/2021
CN 115981052 A 4/2023
(Continued)

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(57) **ABSTRACT**

A backlight module includes stacked light guide plates, a light source module, and a deflecting film. Each light guide plate has a light-incident surface, a light exit surface, a bottom surface, and a light guide structure. The light exit surface and the bottom surface are placed on two opposite sides of the light guide plate and both connected to the light-incident surface, and the light guide structure is formed on the bottom surface. The light-incident surfaces are placed on a light entry side and arranged along a stacking direction. The light source module is disposed on the light entry side and includes light-emitting elements. The light-emitting elements are arranged in a single line along the light-incident surface of each light guide plate, and each light-emitting element has a light emitting surface facing the light-incident surfaces of the plurality of light guide plates.

9 Claims, 8 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

2003/0184990 A1 * 10/2003 Lin B60K 35/22
362/616
2005/0237768 A1 * 10/2005 Matsumoto G02B 6/0036
362/620
2009/0185389 A1 * 7/2009 Tessnow G02B 6/0038
362/516
2021/0239896 A1 8/2021 Tang et al.
2023/0101373 A1 * 3/2023 Kim G02B 6/0036
362/613
2023/0273362 A1 * 8/2023 Yagi G02B 6/0075
362/606
2023/0288753 A1 * 9/2023 Shiau G02F 1/133615
2023/0417978 A1 * 12/2023 Kanbayashi G02B 6/0088

FOREIGN PATENT DOCUMENTS

TW 1592722 B 7/2017
TW 1764775 B 5/2022
TW 202238230 A 10/2022

* cited by examiner

10

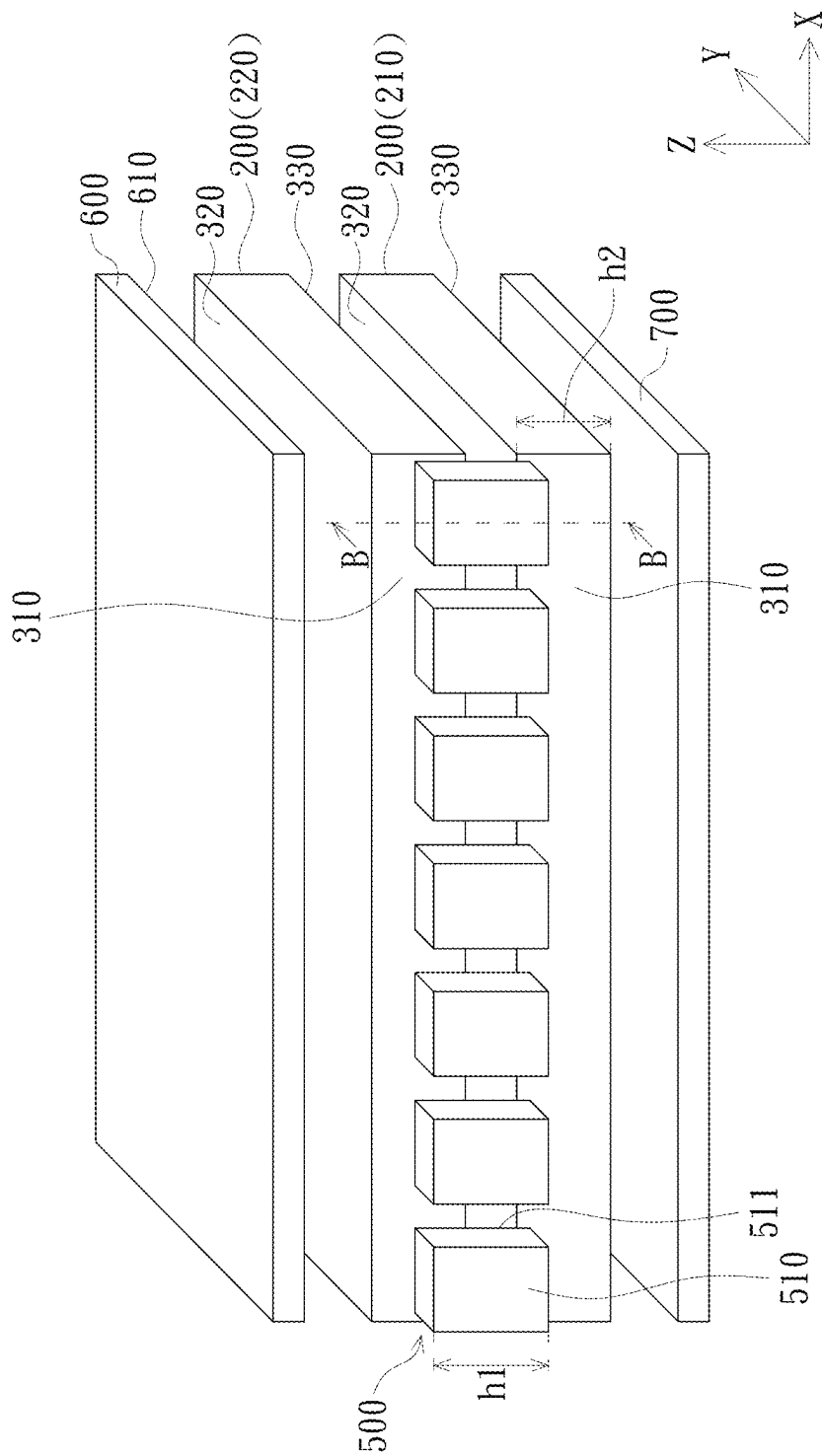


FIG. 1

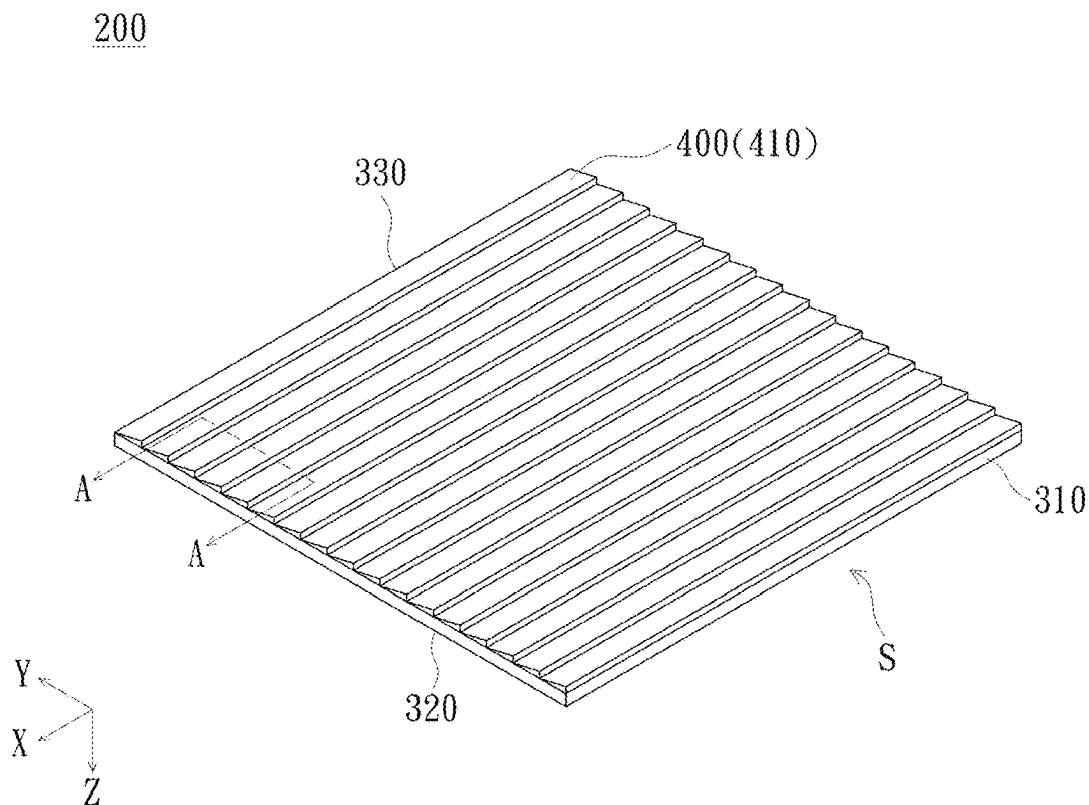


FIG. 2

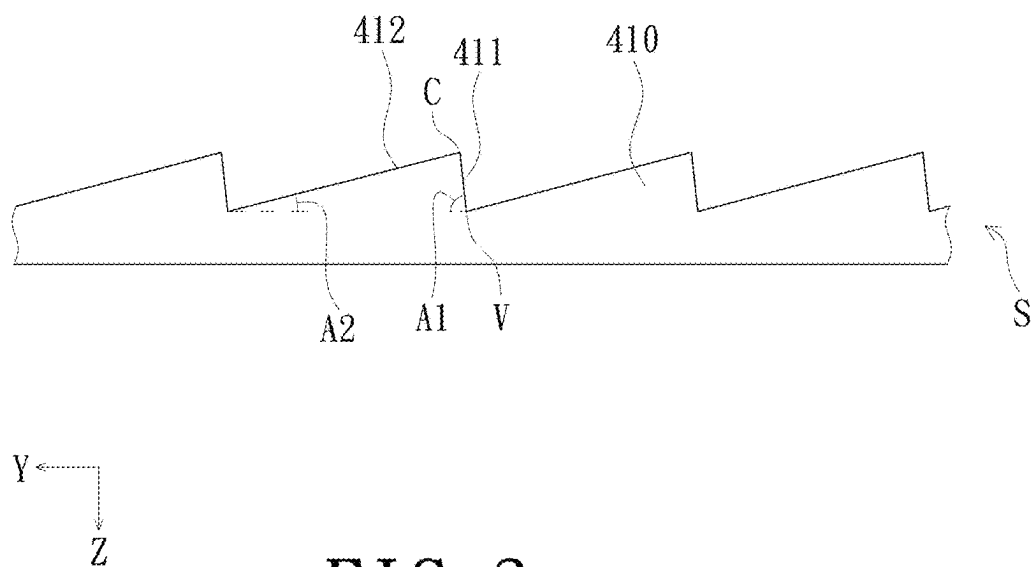


FIG. 3

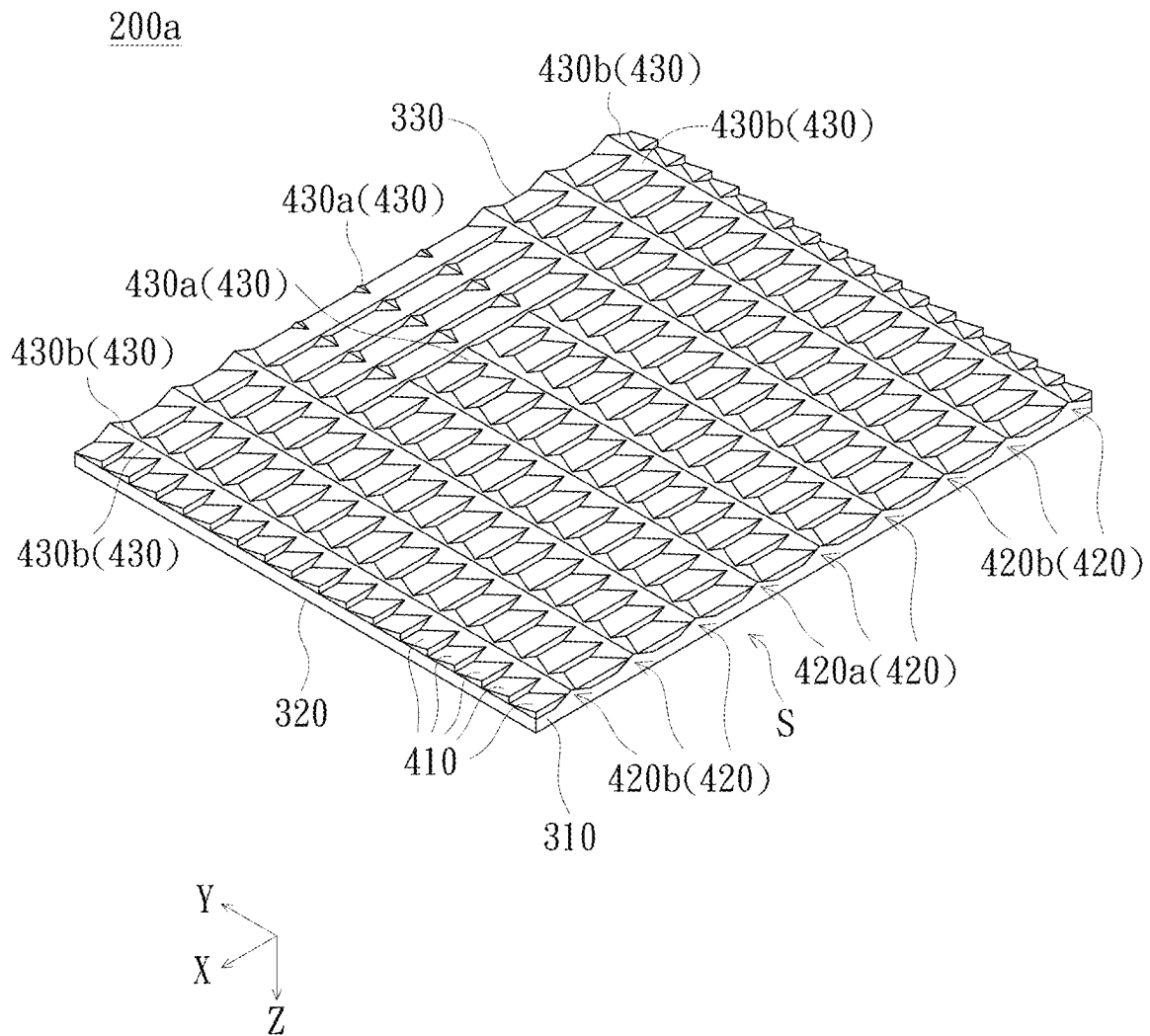


FIG. 4

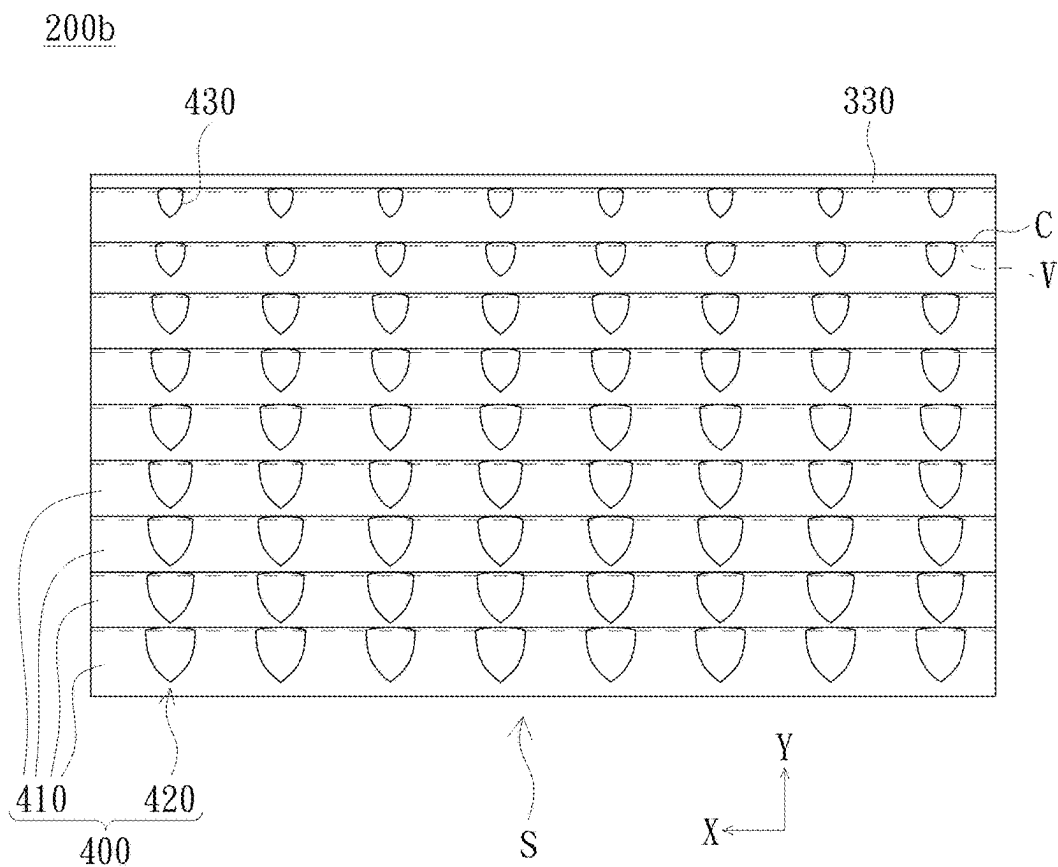


FIG. 5

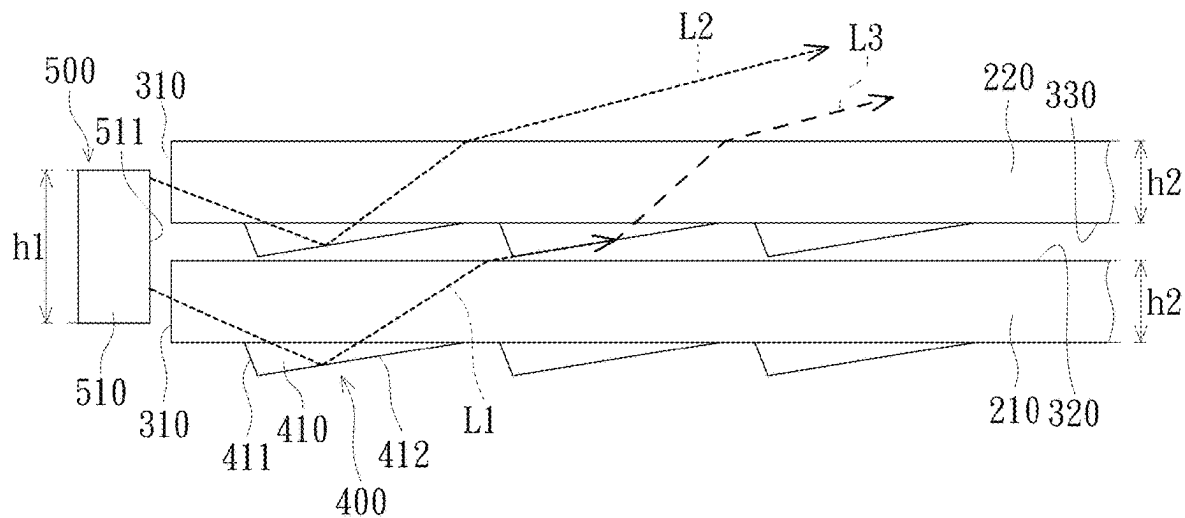


FIG. 6

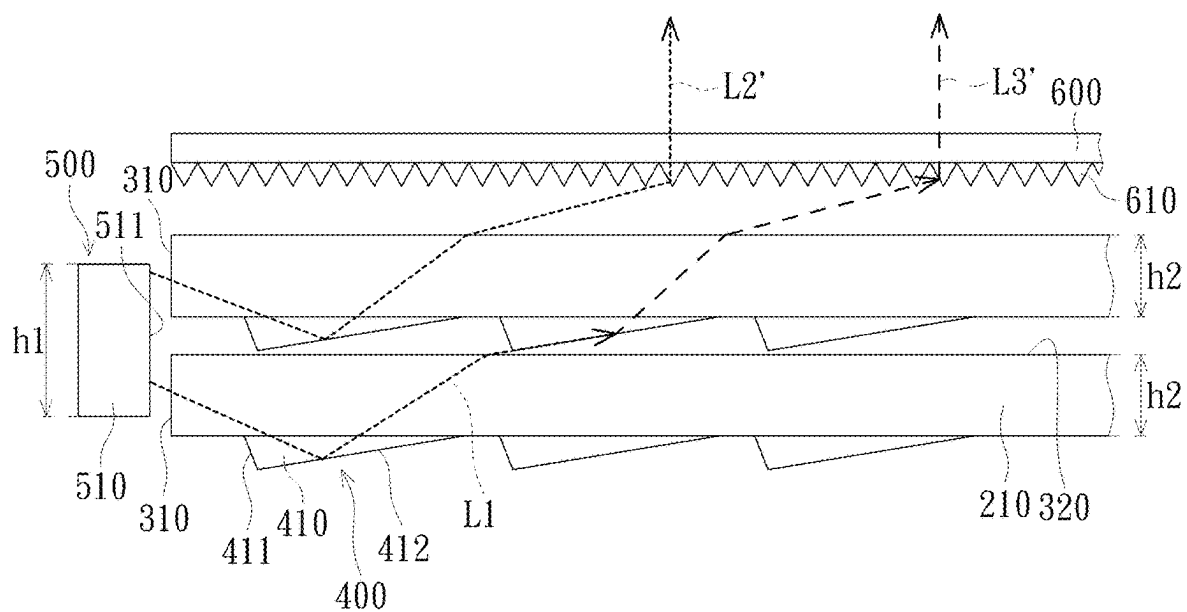


FIG. 7

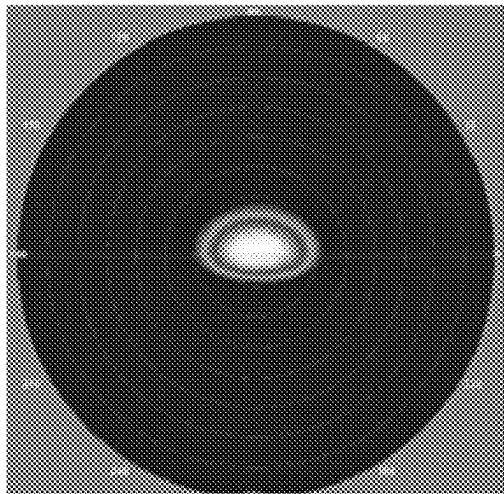


FIG. 8

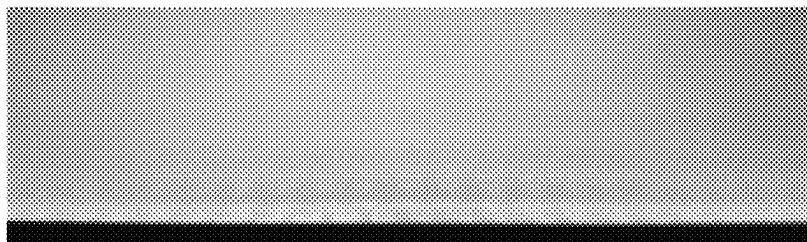


FIG. 9A
(PRIOR ART)

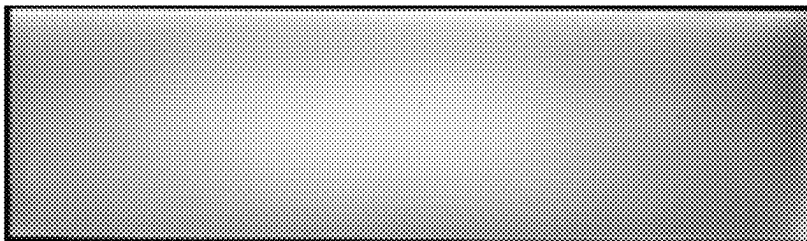


FIG. 9B

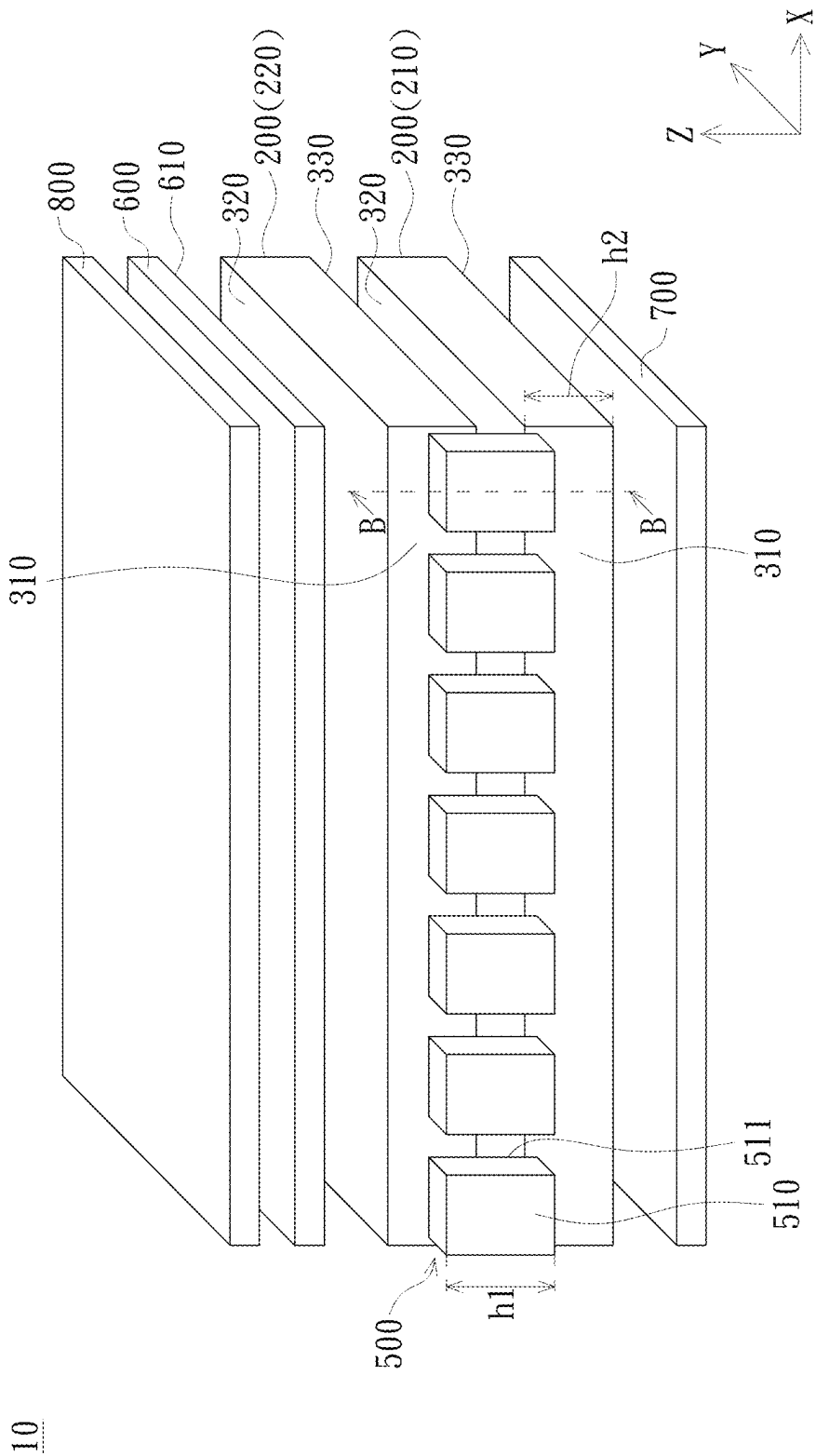


FIG. 10

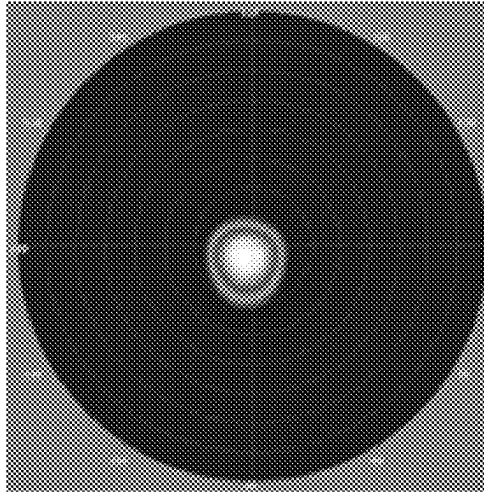


FIG. 11

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BACKLIGHT MODULE INCLUDING LIGHT GUIDE STRUCTURE

FIELD OF THE INVENTION

The present invention relates to a backlight module, and in particular to an anti-peeping backlight module.

BACKGROUND OF THE INVENTION

To meet user privacy and other requirements, conventional backlight modules employ an architecture including a diffusion sheet, a prism sheet, and an anti-peeping sheet, to achieve the effect of viewing angle convergence and anti-peeping. However, this architecture is often faced with problems of low efficiency, insufficient brightness, and excessive power generation, which are caused by a much too big light exit angle of a lower light guide plate, low light utilization due to dispersed energy, and a decrease in brightness after penetration through the anti-peeping sheet by light, and other factors. In addition, this architecture thickens backlight modules and increases costs. To solve problems such as low efficiency and insufficient brightness, light guide plates with a V-cut are being tried out. However, when such light guide plates are applied in automotive displays, industrial control, and other display fields, a projected water ripple phenomenon tends to occur on the light entry side, which affects the visual experience.

SUMMARY OF THE INVENTION

The present invention provides a backlight module. The backlight module has a simplified structure, is capable of reducing costs, and can solve problems such as low light utilization, insufficient brightness, excessive power, etc. In addition, the backlight module has high brightness, can achieve effects of viewing angle convergence and anti-peeping, and can help to eliminate the projected water ripple phenomenon.

The backlight module provided by the present invention includes a plurality of light guide plates, a light source module, and a deflecting film. The plurality of light guide plates are stacked, and each light guide plate includes a light-incident surface, a light exit surface, a bottom surface, and a light guide structure. The light exit surface and the bottom surface are placed on two opposite sides of the light guide plate and both are connected to the light-incident surface, and the light guide structure is formed on the bottom surface. The light-incident surfaces of the plurality of light guide plates are placed on a light entry side and arranged along a stacking direction. The light source module is disposed on the light entry side and includes a plurality of light-emitting elements. The plurality of light-emitting elements are arranged in a single line along the light-incident surface of each light guide plate, and each light-emitting element has a light emitting surface facing the light-incident surfaces of the plurality of light guide plates. The deflecting film is disposed on the plurality of light guide plates.

The present invention employs a plurality of light guide plates, and each light guide plate has a light guide structure. Therefore, the present invention can avoid projected water ripples and can allow a rather small angle range for lights to exit, thereby realizing high light directionality and high light utilization, achieving effects of viewing angle convergence and anti-peeping while maintaining brightness.

Other objectives, features and advantages of the invention will be further understood from the further technological

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features disclosed by the embodiments of the invention wherein there are shown and described preferred embodiments of this invention, simply by way of illustration of modes best suited to carry out the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic three-dimensional diagram of a backlight module according to an embodiment of the present invention;

FIG. 2 is a schematic three-dimensional diagram of a light guide plate according to a first embodiment of the present invention;

FIG. 3 is a partial sectional view of a light guide plate according to the first embodiment of the present invention;

FIG. 4 is a schematic three-dimensional diagram of a light guide plate according to a second embodiment of the present invention;

FIG. 5 is a schematic bottom view of a light guide plate according to a third embodiment of the present invention;

FIG. 6 is a schematic sectional view of the embodiment in FIG. 1;

FIG. 7 is another schematic sectional view of the embodiment in FIG. 1;

FIG. 8 is a light field distribution diagram of an embodiment of the present invention;

FIG. 9A is a schematic image diagram of a light entry side of an architecture of a conventional backlight module;

FIG. 9B is a schematic image diagram of a light entry side of a backlight module according to an embodiment of the present invention;

FIG. 10 is a schematic three-dimensional diagram of a backlight module according to another embodiment of the present invention; and

FIG. 11 is a light field distribution diagram of another embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

The foregoing and other technical contents and other features and advantages of the present invention will be clearly presented from the following detailed description of a preferred embodiment in cooperation with the accompanying drawings. Directional terms mentioned in the following examples, for example, upper, lower, left, right, front, back, top or bottom, are only used to describe directions referring to the attached drawings. Therefore, the directional terms used are for illustration and not for limitation. In addition, terms such as "first" and "second" involved in the description or claims are merely used for naming the elements or distinguishing different embodiments or ranges rather than limiting the upper limit or lower limit of the quantity of the elements.

FIG. 1 is a schematic three-dimensional diagram of a backlight module according to an embodiment of the present invention. As shown in FIG. 1, in this embodiment of the present invention, the backlight module 10 includes a plurality of light guide plates 200, a light source module 500, and a deflecting film 600. The plurality of light guide plates 200 are stacked and each light guide plate 200 has a light-incident surface 310, a light exit surface 320, a bottom surface 330, and a light guide structure 400 (will be described in the following). The light exit surface 320 and the bottom surface 330 are placed on two opposite sides of the light guide plate 200 and both connected to the light-incident surface 310. The light guide structure 400 is formed

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on the bottom surface **330**. The light guide structure **400** may be in any shape, for example, a V-cut, and allow the lights traveling in the light guide plate **200** to exit from the light exit surface **320**. In this embodiment of the present invention, the light guide structure **400** is further used to render the reflected lights more directional.

Further, the plurality of light guide plates **200** are arranged along a stacking direction **Z**. The bottom surface **330** of the upper light guide plate **200** faces the light exit surface **320** of the lower light guide plate **200**, and the light-incident surfaces **310** are both placed on a light entry side **S**. Preferably, the plurality of light-incident surfaces **310** are substantially arranged along the stacking direction **Z** and aligned. The light source module **500** is disposed on the light entry side **S** and includes a plurality of light-emitting elements **510**. Each light-emitting element **510** has a light emitting surface **511** facing the light-incident surfaces **310** of the plurality of light guide plates **200**, and is suitable for emitting light to the plurality of light guide plates **200** through the plurality of light-incident surfaces **310**.

In this embodiment of the present invention, the plurality of light-emitting elements **510** are arranged in a line. In a preferred embodiment, the plurality of light-emitting elements **510** are arranged in a single line along the light-incident surface **310** of the light guide plate **200** in a direction **X**. The light-emitting element **510** has a height **h1** in the stacking direction **Z**, and the height **h1** is preferably greater than a height **h2** of a single light guide plate **200** in the stacking direction **Z**. In other words, the light-emitting element **510** in this embodiment of the present invention may have a large dimension, and the height **h1** is greater than that of the light guide plate **200**. In a preferred embodiment of the present invention, the light-emitting element **510** has a long service life, and is therefore suitable for an apparatus with a specific backlight requirement. In various embodiments of the present invention, the backlight module **10** may be used in automotive displays, industrial control, and other display fields. The light-emitting elements **510** can have a large dimension and a long service life, and can thus provide backlight for tens of thousands of hours.

FIG. 2 is a schematic three-dimensional diagram of a light guide plate according to an embodiment of the present invention, and FIG. 3 is a schematic sectional view along a section line A-A in FIG. 2. As shown in FIG. 2, the light guide structure **400** includes a plurality of prisms **410** arranged along a direction **Y** away from the light entry side **S**. The prism **410** can control the light exit angle at which the light exits from the light exit surface **320**. As shown in FIG. 3, each prism **410** has a first surface **411** and a second surface **412**. The first surface **411** faces the light entry side **S**, the second surface **412** faces away from the light entry side **S**, and the first surface **411** and the second surface **412** border each other along an edge line **C** of the prism **410**. Sections of the prisms **410** shown by FIGS. 2 and 3 are roughly triangular and spaced apart from each other at fixed intervals, but the present invention is not limited thereto. In addition, the first surface **411** and the second surface **412** may further border each other on a curved surface. In other words, the prism **410** has an edge portion (not shown), and the edge portion may have no obvious edge line **C**. The section of the edge portion may be, for example, a fillet with a radius of **R**. In a preferred embodiment of the present invention, the plurality of prisms **410** can allow a rather small angle range for lights to exit.

The first surface **411** has a first angle **A1** with respect to the bottom surface **330**, and the second surface **412** has a second angle **A2** with respect to the bottom surface **330**. The

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first angle **A1** may not be equal to the second angle **A2**. In various embodiments of the present invention, the first angle **A1** is greater than the second angle **A2**, and the second angle **A2** is preferably smaller than 15 degrees, more preferably smaller than 3 degrees, so as to allow a specific angle for lights to exit. For example, the light exit angle (which is an angle with respect to a normal of the light exit surface **320**) may be 70 to 80 degrees. In addition, in this embodiment of the present invention, if the second angle **A2** is smaller than three degrees, lights can be further allowed a smaller angle range to exit and thus become more directional.

FIG. 4 is a schematic three-dimensional diagram of a light guide plate according to another embodiment of the present invention. As shown in FIG. 4, the light guide structure **400** may further include a plurality of microstructure groups **420**, each including a plurality of microstructures **430** and arranged between two adjacent prisms **410**. The microstructure **430** can improve the uniformity of surface light sources, thereby avoiding bright or dark ripples. The plurality of microstructures **430** in each microstructure group **420** may be partially or completely connected or not connected to each other. Additionally, the plurality of microstructures **430** in each microstructure group **420** may be arranged along a preset path. For example, as shown by FIG. 4, the plurality of microstructure groups **420** further include first microstructure groups **420a** and second microstructure groups **420b**. The plurality of first microstructures **430a** of the first microstructure groups **420a** are arranged along a linear path and partially connected, and the plurality of second microstructures **430b** of the second microstructure groups **420b** are completely connected. In another embodiment, the plurality of microstructures **430** may alternatively be arranged along, for example, a zigzag path, thereby forming a staggered pattern.

The microstructure **430** shown in FIG. 4 is a raised bulge block, but the microstructure **430** may be a concaved groove in another embodiment. The microstructure **430** may be pyramid-shaped. For example, some of the microstructures **430a** in FIG. 4 are pyramid-shaped with a planar side surface. However, the microstructure **430** may have a curved surface in another embodiment. FIG. 5 is a schematic bottom view of a light guide plate according to another embodiment of the present invention. As shown in FIG. 5, the light guide structure **400** of the light guide plate **200b** includes a plurality of prisms **410** and a plurality of microstructure groups **420**. On the bottom surface **330**, a solid line is used to represent the edge line **C** of the prism **410**, and a dotted line is used to represent a valley line **V** of the prism **410**. The light guide plate in this embodiment and the foregoing light guide plate **200a** differ in that the surface of the microstructure **430** is a curved surface, and the shape of the section thereof may have a curved line.

In various embodiments of the present invention, the number of light guide plates **200** may be two, which include, for example, a first light guide plate **210** and a second light guide plate **220**. FIG. 6 is a schematic sectional view along a section line B-B in FIG. 1. As shown in FIG. 6, the first light guide plate **210** and the second light guide plate **220** are stacked, the second light guide plate **220** is placed on a light exit surface **320** side of the first light guide plate **210**, and the light emitting surface **511** of the light-emitting element **510** of the light source module **500** faces the light-incident surfaces **310** of the first light guide plate **210** and the second light guide plate **220**. As described above, the first light guide plate **210** and the second light guide plate **220** both have a height of **h2**. In this embodiment of the present invention, **h2** is smaller than **h1**, and a sum of the heights of

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the plurality of light guide plates **200** (e.g. a sum of the heights of the first light guide plate **210** and the second light guide plate **220**, $h2+h2$) is preferably greater than the height $h1$. The fact that $h2+h2 \geq h1$ helps ensure that the light emitting surface **511** can face the light-incident surfaces **310** of the plurality of light guide plates **200** to the maximum extent, such that the light source can effectively enter the light guide plates **200** so as to generate a surface light source. The number of light guide plates **200** may alternatively be more than two, for example, three, four, or five, and preferably no more than five. Regardless of the number, the sum of the heights of the plurality of light guide plates **200** is preferably greater than the height $h1$, and the light emitting surface **511** of the light-emitting element **510** preferably faces the light-incident surfaces **310** of the plurality of light guide plates **200**.

In various embodiments of the present invention, the first light guide plate **210** and the second light guide plate **220** are the same and have the same light guide structure **400**. As shown in FIGS. **2** and **6**, because the first light guide plate **210** and the second light guide plate **220** each have a plurality of prisms **410**, lights can always exit from each light guide plate **200** within a specific angle range regardless of whether the light source enters the first light guide plate **210** or the second light guide plate **220**. That is, the light path **L1** and the light path **L2** are substantially within the same angle range. Since the prism **410** has a second angle $A2$ that is small and nearly horizontal, the amount of deflection can still be limited to a small angle as lights exiting from the first light guide plate **210** enter the second light guide plate **220** after passing through the light guide structure **400** from the bottom surface **330** side of the second light guide plate **220**, even in the case where lights are deflected in a manner shown by a light path **L3**. In various embodiments of the present invention, the amount of deflection may be smaller than three degrees.

In other embodiments of the present invention, the plurality of light guide plates **200** may also be different, for example, having different microstructure groups **420**. However, although the plurality of light guide plates **200** may be different from each other, they are all highly directional, and the sum of the heights of the plurality of light guide plates **200** is greater than or equal to the height $h1$. Even if the plurality of light guide plates **200** are different from each other, the different light guide plates **200** can still preferably allow the same specific angle within the same small angle range for lights to exit, and the amount of deflection of exiting lights, which is caused by the stacking of a plurality of light guide plates **200**, can be limited to a small angle.

As shown in FIGS. **1** and **7**, the deflecting film **600** of the backlight module **10** is disposed on the plurality of light guide plates **200**. The deflecting film **600** has a prism surface **610**, and the prism surface **610** faces the light exit surface **320** of the light guide plate **200**, for example, the light exit surface **320** of the second light guide plate **220**. In a preferred embodiment of the present invention, as shown by light paths **L2'** and **L3'**, the deflecting film **600** can guide the light exiting from the plurality of light guide plates **200** towards a normal direction or a nearby range thereof. Because the plurality of light guide plates **200** can allow a small angle range for lights to exit, and the amount of deflection of the exiting lights is small, the deflecting film **600** can effectively guide the exiting lights towards a normal direction or a small nearby range thereof.

FIG. **8** is a light field distribution diagram of a backlight module according to the present invention. As shown in FIG. **8**, the light is concentrated in the center and its nearby area,

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indicating that the exiting lights are guided towards a normal direction and a nearby range thereof in this embodiment of the present invention. Therefore, the effects of viewing angle convergence and anti-peeping can be achieved. In addition, when the backlight module **10** in the embodiments of the present invention is used in automotive displays, industrial control, and other display fields to replace the thick light guide plates currently employed in these display fields, improvements in terms of reduced projected water ripple phenomenon can be observed, as shown by FIG. **9A** (prior art) and **9B** (present invention embodiment).

As shown in FIG. **1**, the backlight module **10** in the embodiments of the present invention may further include a reflecting sheet **700** configured to reflect lights exiting from the bottom surface **330**, so as to increase light utilization. In addition, as shown in FIG. **10**, the backlight module **10** of this embodiment of the present invention may further include an anti-peeping sheet **800**. The anti-peeping sheet **800** can reduce the energy of lights exiting at a large viewing angle, thereby further enhancing the anti-peeping effect. FIG. **11** is a light field distribution diagram according to the embodiment shown in FIG. **10**. Those skilled in the art of the present invention can determine whether to use the anti-peeping sheet **800** based on specific brightness requirements and anti-peeping needs.

The backlight module **10** in the embodiments of the present invention is highly directional and can allow a smaller angle range for lights to exit, and therefore ensures high light utilization, and is capable of achieving the effects of viewing angle convergence and anti-peeping while maintaining brightness. Although the anti-peeping sheet is used to improve the anti-peeping effect, the brightness can still be maintained at a certain level.

While the invention has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

What is claimed is:

1. A backlight module, comprising:

a plurality of light guide plates, wherein the plurality of light guide plates are stacked, each of the light guide plates comprises a light-incident surface, a light exit surface, a bottom surface, and a light guide structure; wherein:

the light exit surface and the bottom surface are located on two opposite sides of the light guide plate and both connected to the light-incident surface;

the light guide structure is formed on the bottom surface, and comprises a plurality of prisms; the plurality of prisms are arranged along a direction away from the light entry side, wherein each of the prisms has a first surface and a second surface, and the first surface and the second surface border each other along an edge line of the prism; and

the light-incident surfaces of the plurality of light guide plates are placed on a light entry side and arranged along a stacking direction;

a light source module, disposed on the light entry side and comprising a plurality of light-emitting elements, wherein the plurality of light-emitting elements are arranged in a single line along the light-incident surface of the plurality of light guide plates, and each of the

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light-emitting elements has a light emitting surface facing the light-incident surfaces of the plurality of light guide plates; and
a deflecting film, disposed on the plurality of light guide plates.

2. The backlight module according to claim 1, wherein the light guide structure further comprises a plurality of microstructure groups, and each of the microstructure groups comprises a plurality of microstructures and is arranged between two adjacent prisms.

3. The backlight module according to claim 1, wherein the first surface has a first angle with respect to the bottom surface, the second surface has a second angle with respect to the bottom surface, and the first angle is not equal to the second angle.

4. The backlight module according to claim 3, wherein the first angle is greater than the second angle, and wherein the second angle is smaller than 15 degrees.

5. The backlight module according to claim 4, wherein the first surface faces the light entry side, the second surface faces away from the light entry side, and the second angle is smaller than 3 degrees.

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6. The backlight module according to claim 1, wherein the plurality of light-emitting elements are suitable for emitting light to the plurality of light guide plates through the plurality of light-incident surfaces, the plurality of light guide plates are suitable for light to exit at a light exit angle, and the light exit angle is greater than 70 degrees.

7. The backlight module according to claim 1, wherein a height of the light-incident surface of the light guide plate is smaller than a height of the light emitting surface of the light-emitting element along the stacking direction.

8. The backlight module according to claim 7, wherein the plurality of light guide plates further comprise a first light guide plate and a second light guide plate, the second light guide plate is disposed on a light exit surface side of the first light guide plate, and the height of the plurality of light guide plates on the light-incident surface side is parallel to the stacking direction and is greater than or equal to the height of the light emitting surface.

9. The backlight module according to claim 8, wherein the deflecting film further has a prism surface, and the prism surface of the deflecting film faces the light exit surface of the second light guide plate.

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